6.4.1 What are the limitations of AWS Lambda?

Executing a Lambda function cannot exceed 15 minutes. This means the problem you are solving with your function needs to be small enough to fit into the 900-second limit. It is probably not possible to download 10 GB of data from S3, process the data, and insert parts of the data into a database within a single invocation of a Lambda function. But even if your use case fits into the 900-second constraint, make sure that it will fit under all circumstances. Here's a short anecdote from one of our first serverless projects: we built a serverless application that preprocessed analytics data from news sites. The Lambda functions typically processed the data within less than 180 seconds. But when the 2017 US elections came, the volume of the analytics data exploded in a way no one expected. Our Lambda functions were no longer able to complete within 300 seconds—which was the maximum back then. It was a show-stopper for our serverless approach.

AWS Lambda provisions and manages the resources needed to run your function. A new execution context is created in the background every time you deploy a new version of your code, go a long time without any invocations, or when the number of concurrent invocations increases. Starting a new execution context requires AWS Lambda to download your code, initialize a runtime environment, and load your code. This process is called a *cold start*. Depending on the size of your deployment package, the runtime environment, and your configuration, a cold start could take from a few milliseconds to a few seconds.